AI Agent

2026 Fall

소프트웨어융합과

조상구 교수

경복대학교 2025년도 Capstone Design

1. 한국어 음성 및 제스쳐 인식 제어 드론
2. 자율주행 자동차
3. 스마트 홈 (얼굴인식 기반 출입 제어 시스템)
4. IoT 낙상감지 시스템
5. AI Tutor

목차

1. AI는 왜 똑똑하지?

2. AI는 어떻게 학습하지?

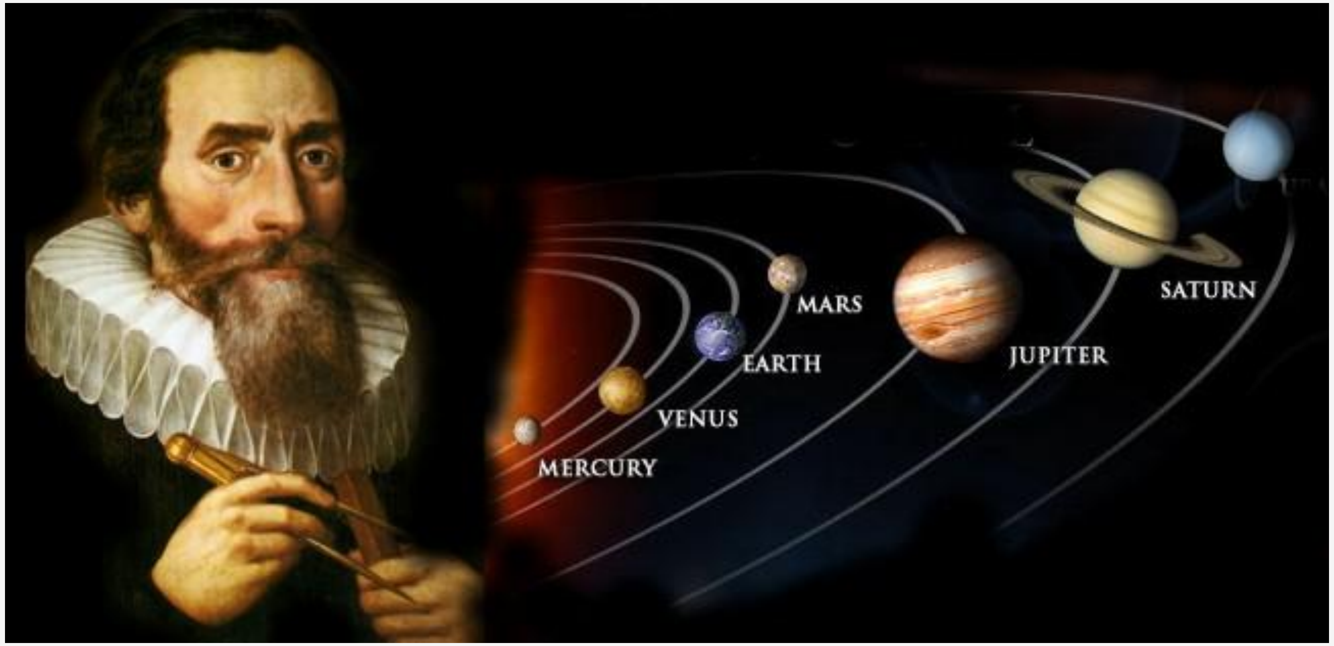
3. 생성형 AI

4. AI agent

사람이 규칙을 직접 찾기

머신러닝이전 시대에는 관측된 데이터에 존재하는 규칙성(Patterns)을 사람이 직접 궁리하고 발견하여 새로운 데이터를 예측 (귀납법)

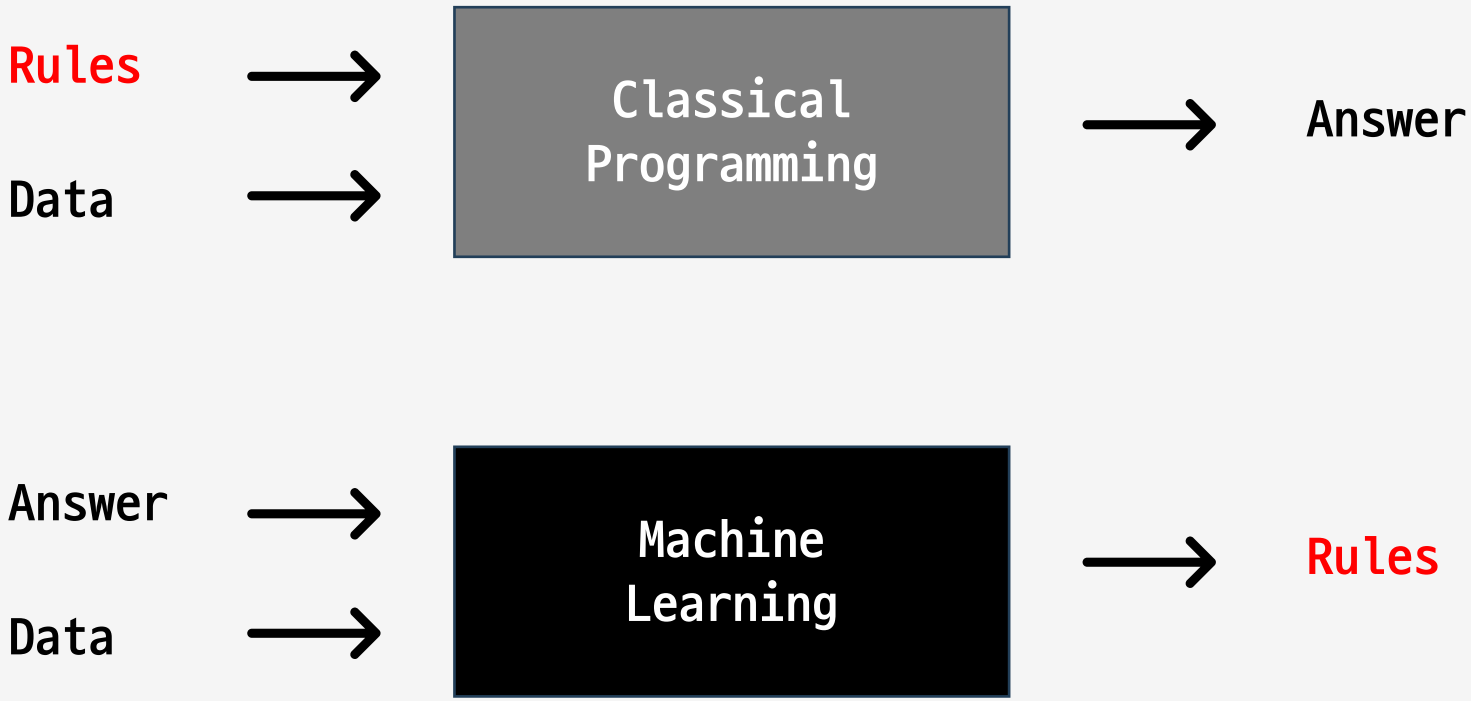
방대한 데이터들을 직접
수년간 궁리하고
계산(Compute)한 끝에,
행성의 공전궤도는 원이
아닌 '타원'이라는
사실(규칙성, 패턴)을 발견



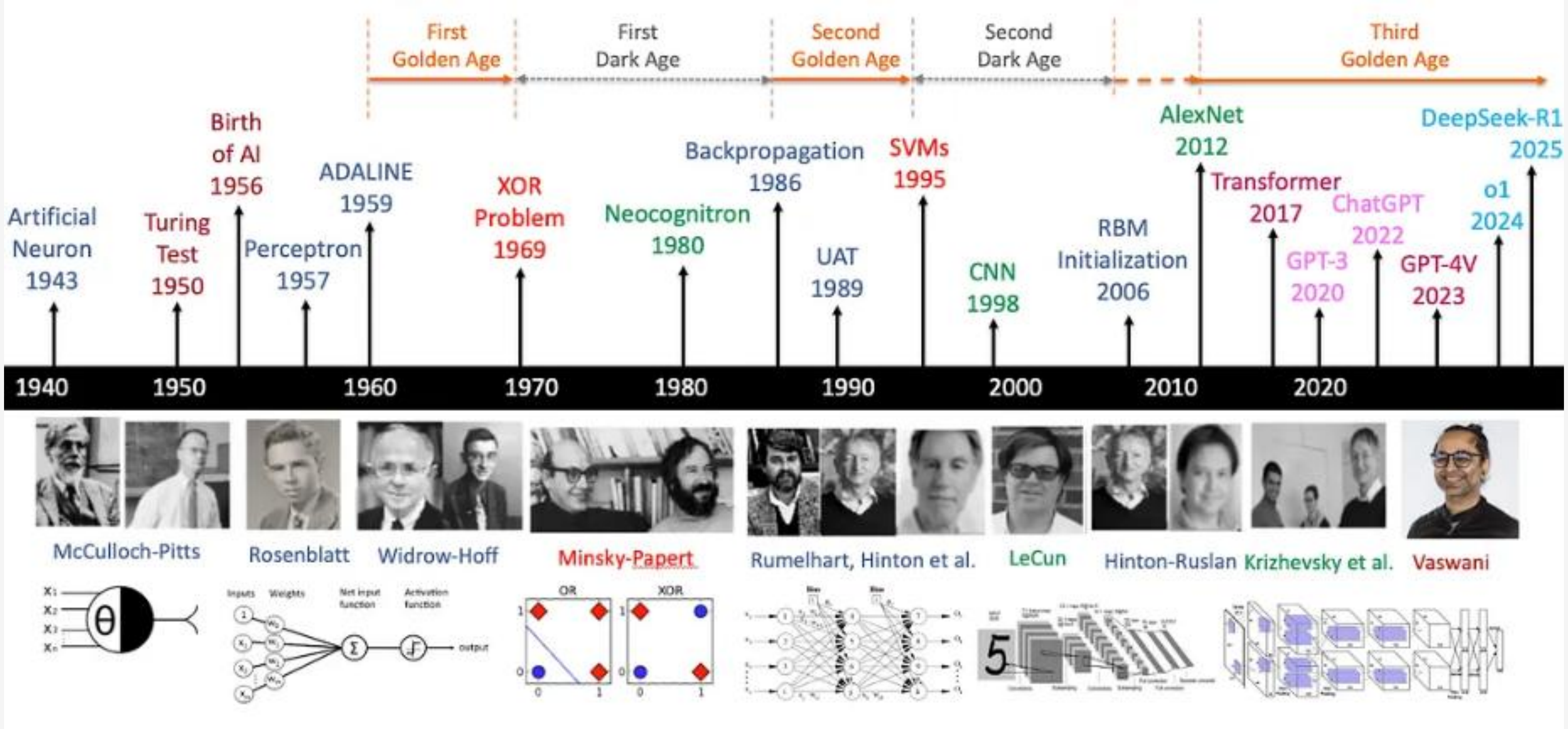
머신러닝

과거 데이터에 존재하는 규칙성(Patterns)을 컴퓨터가 발견하게 하여 새로운 데이터를 예측하는 것이

머신러닝(Machine Learning) → New Paradigm



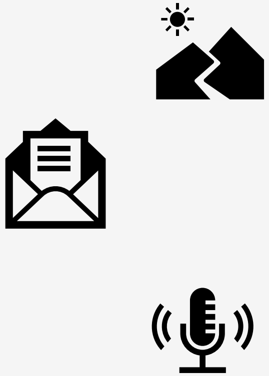
Deep Learning Timeline

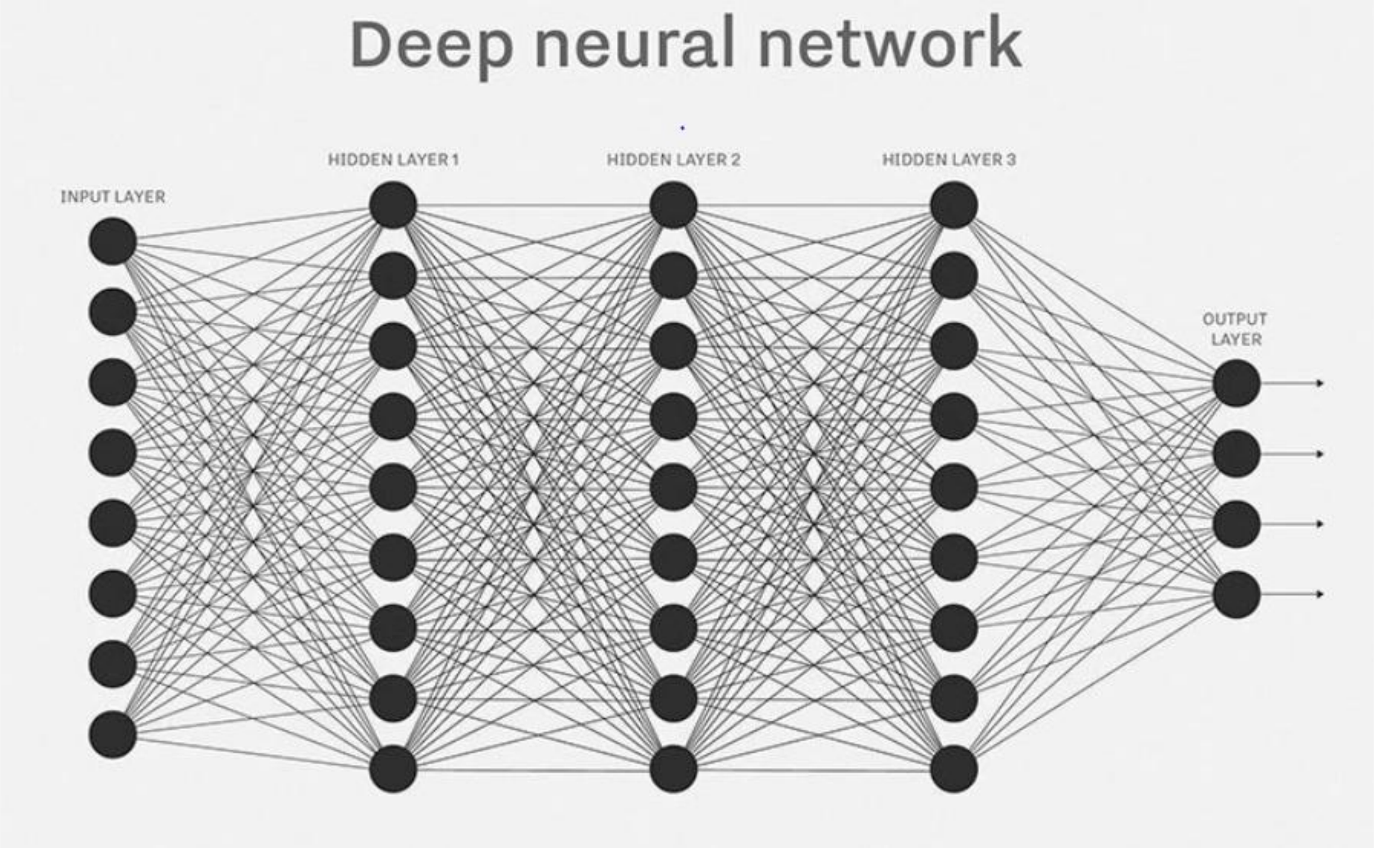


A-brief-history-of-ai-with-deep-learning

AI가 똑똑한 이유?

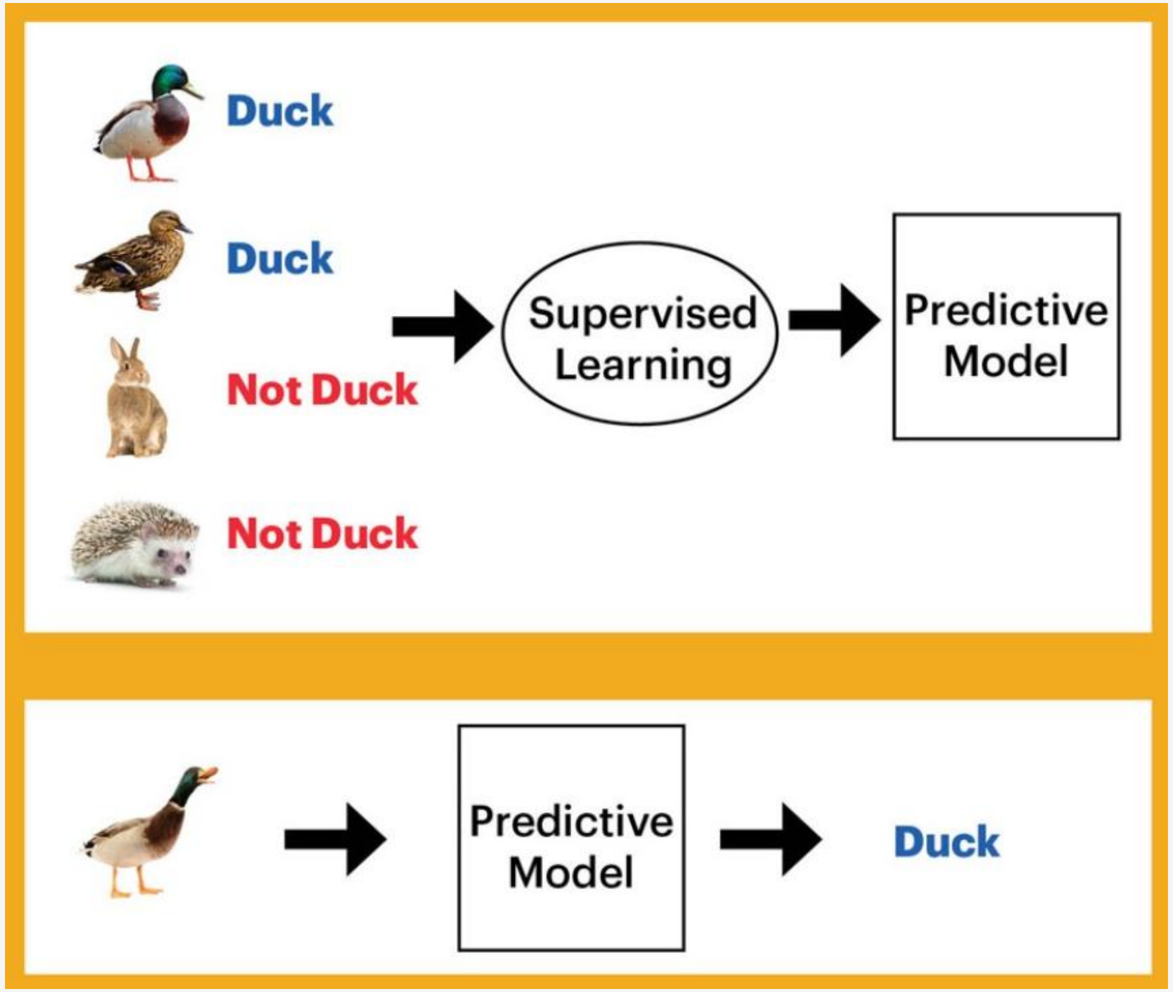
AI가 똑똑한 이유는 엄청난 양의 데이터(이미지, 텍스트, 사운드, 동영상)를 초고속 컴퓨터(학습 능력)로 밤낮없이 읽으면서, 입력받은 모든 데이터의 패턴을 찾아냈기 때문(수능 만점자가 기출문제를 완벽하게 마스터해서 어떤 변형 문제가 나와도 맞히는 것과 비슷)

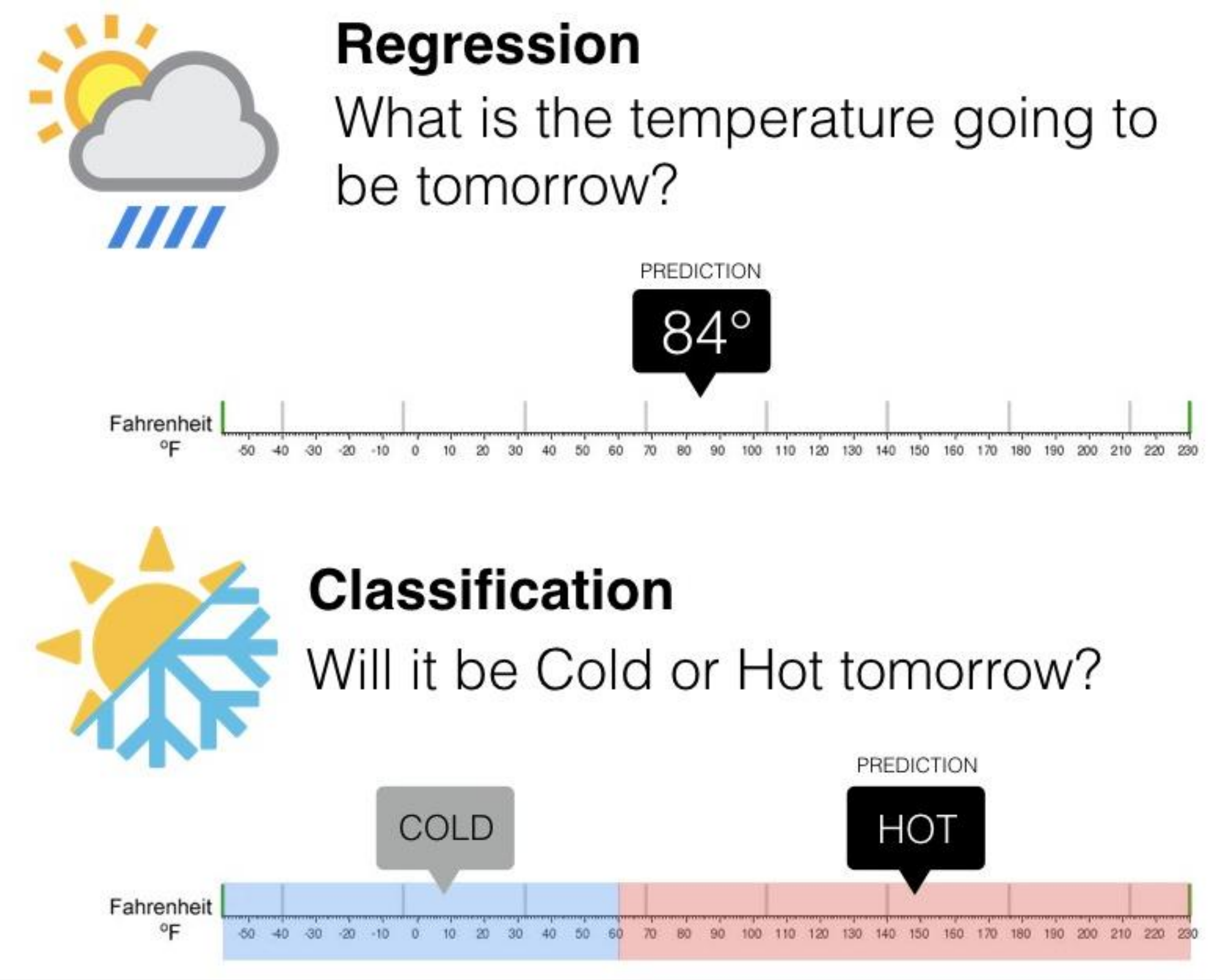




- Big Data
- Computing power
- Algorithm

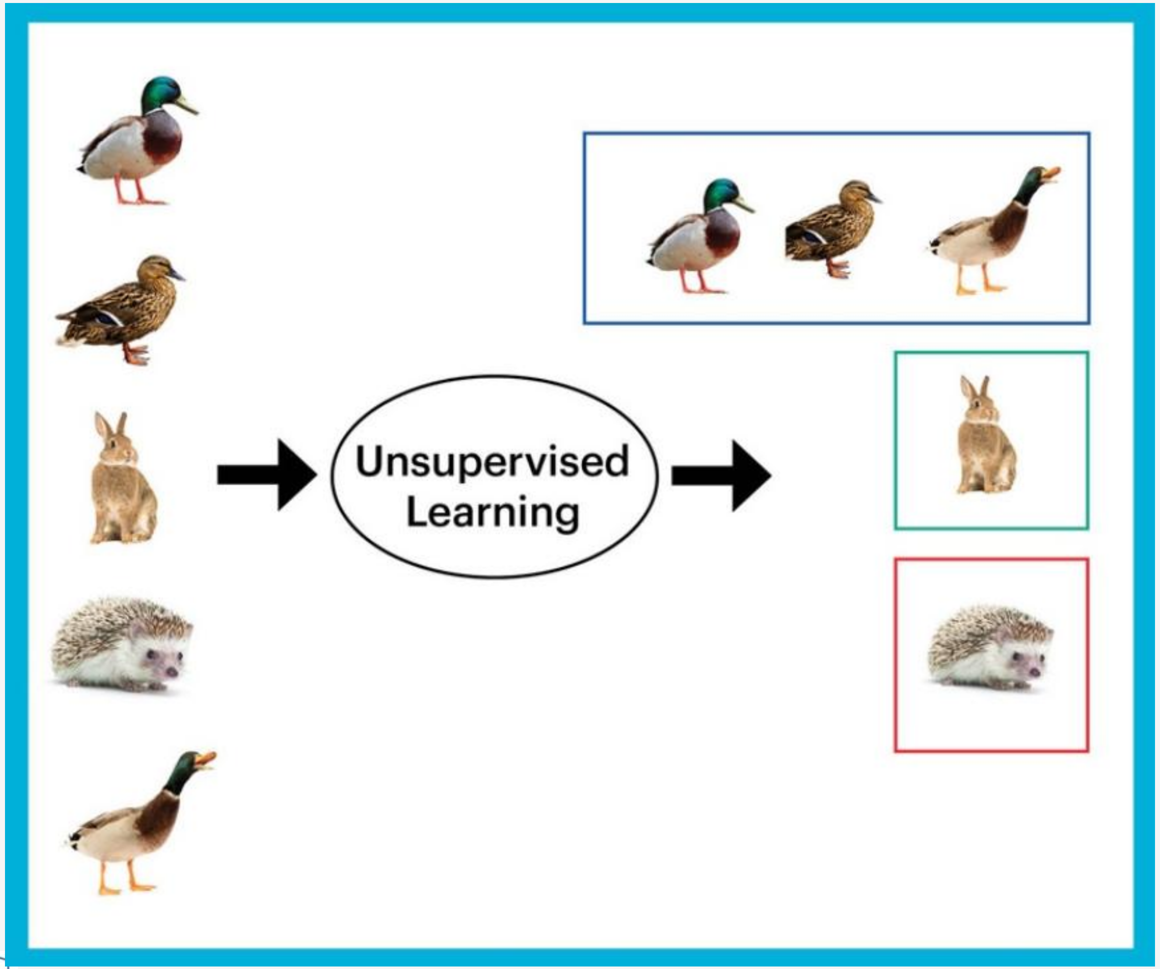
Supervised Learning(지도학습)

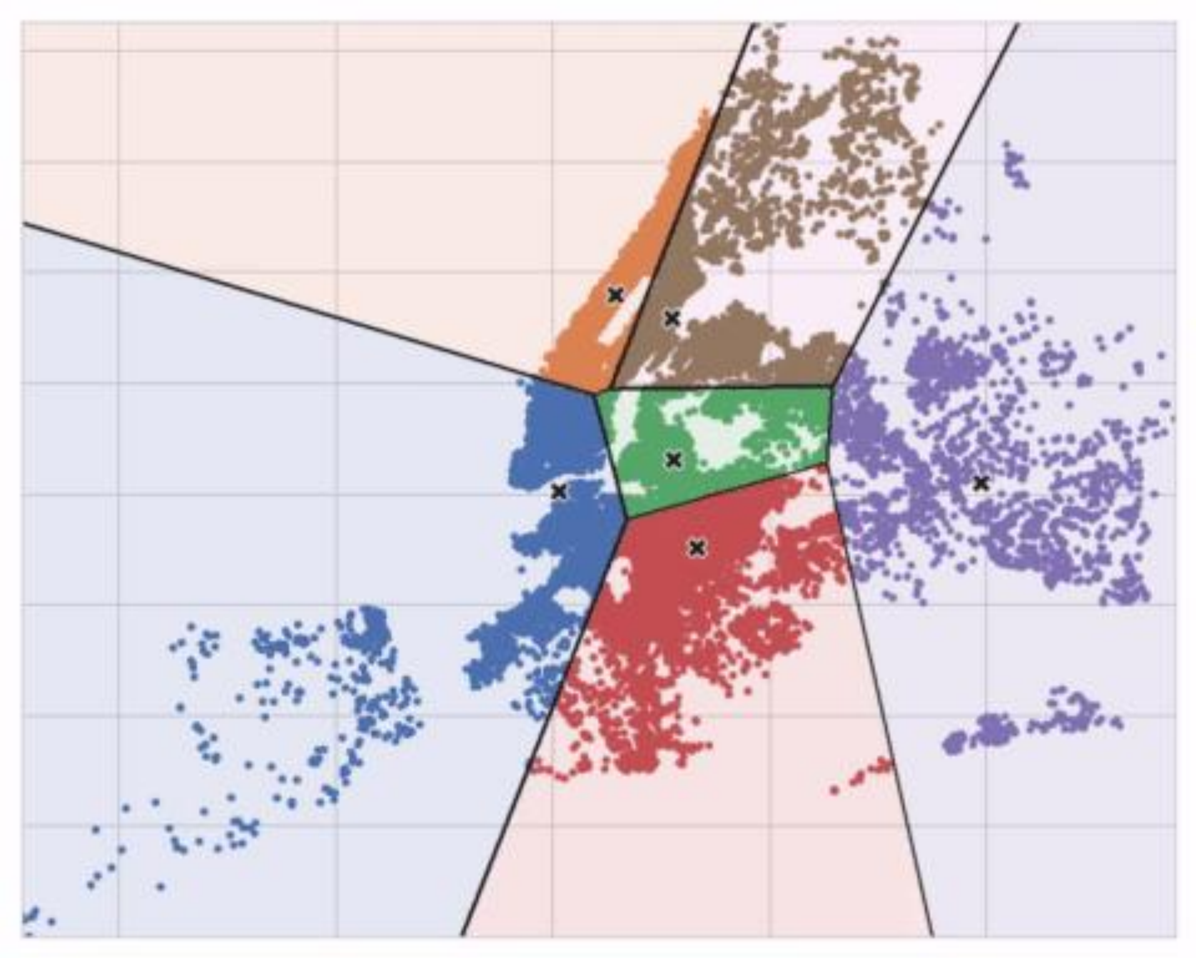




Unsupervised Learning(비지도학습)

Cluster(군집)





목차

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2. AI는 어떻게 학습하지?
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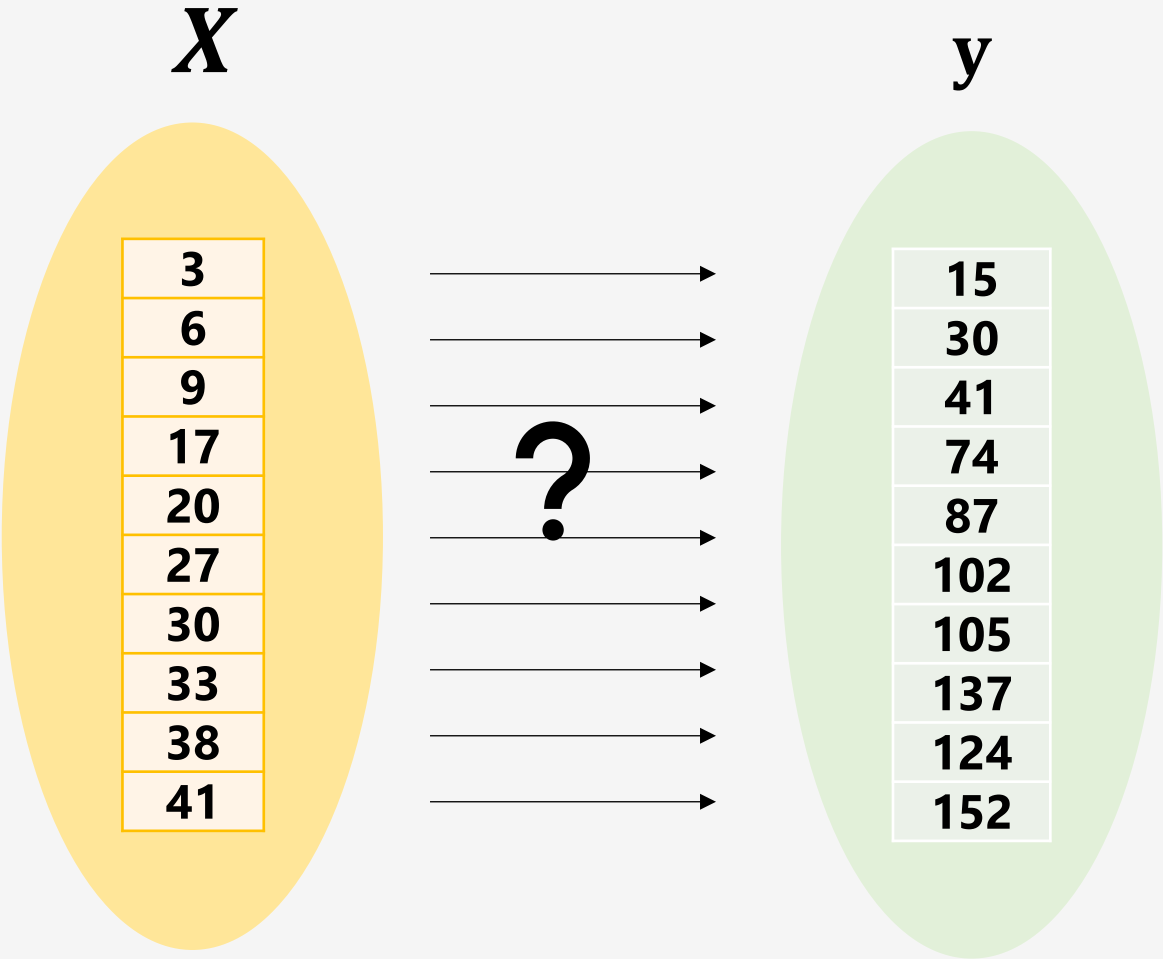
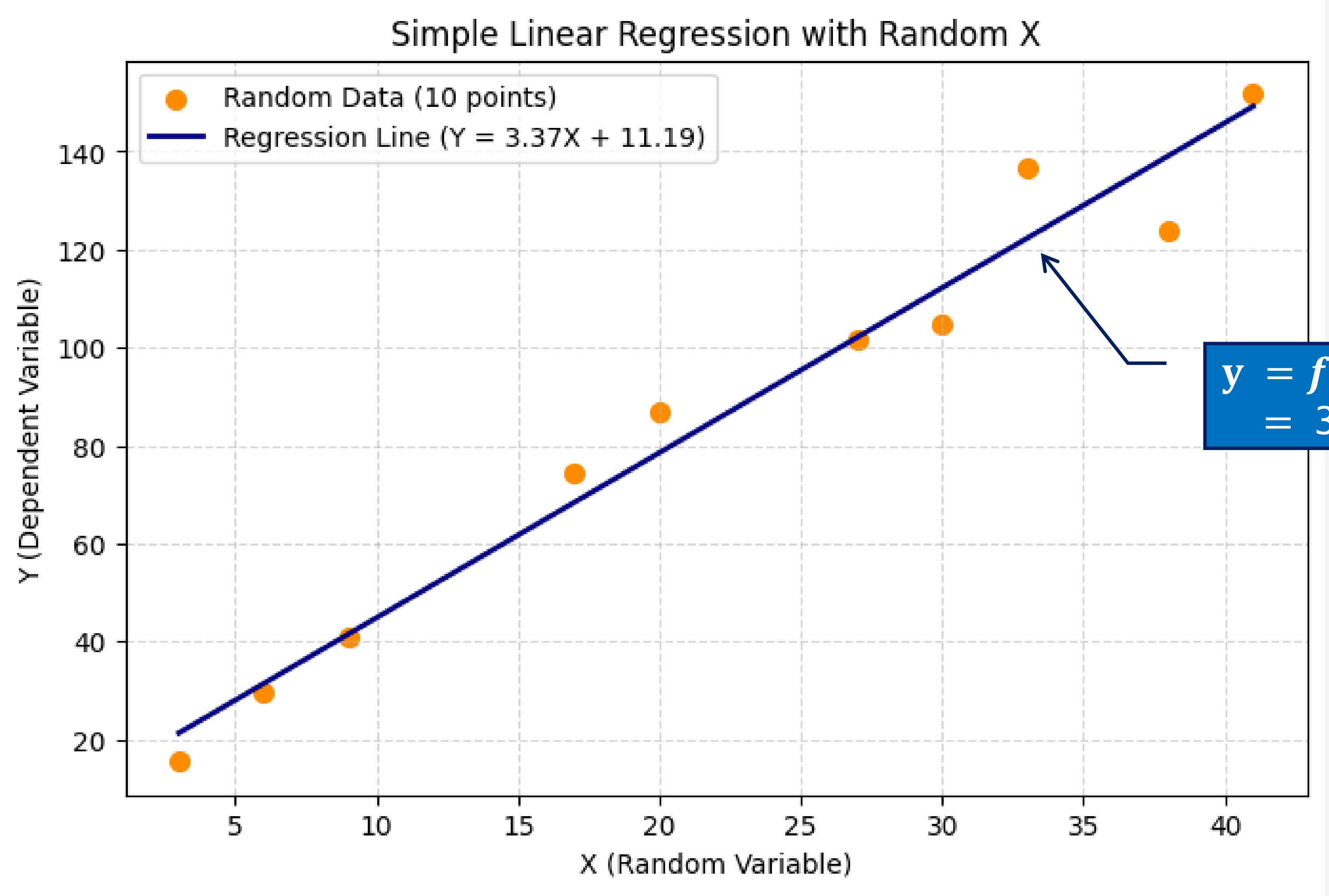
AI는 어떻게 학습하지?



$$\mathbf{y} = f(X)$$

→ find $f($ $)$

AI는 어떻게 학습하지?



$$\mathbf{y} = \boldsymbol{f}(\boldsymbol{X}) = 3.37\boldsymbol{X} + 11.19$$

AI는 어떻게 학습하지?

X **y**

wired.com

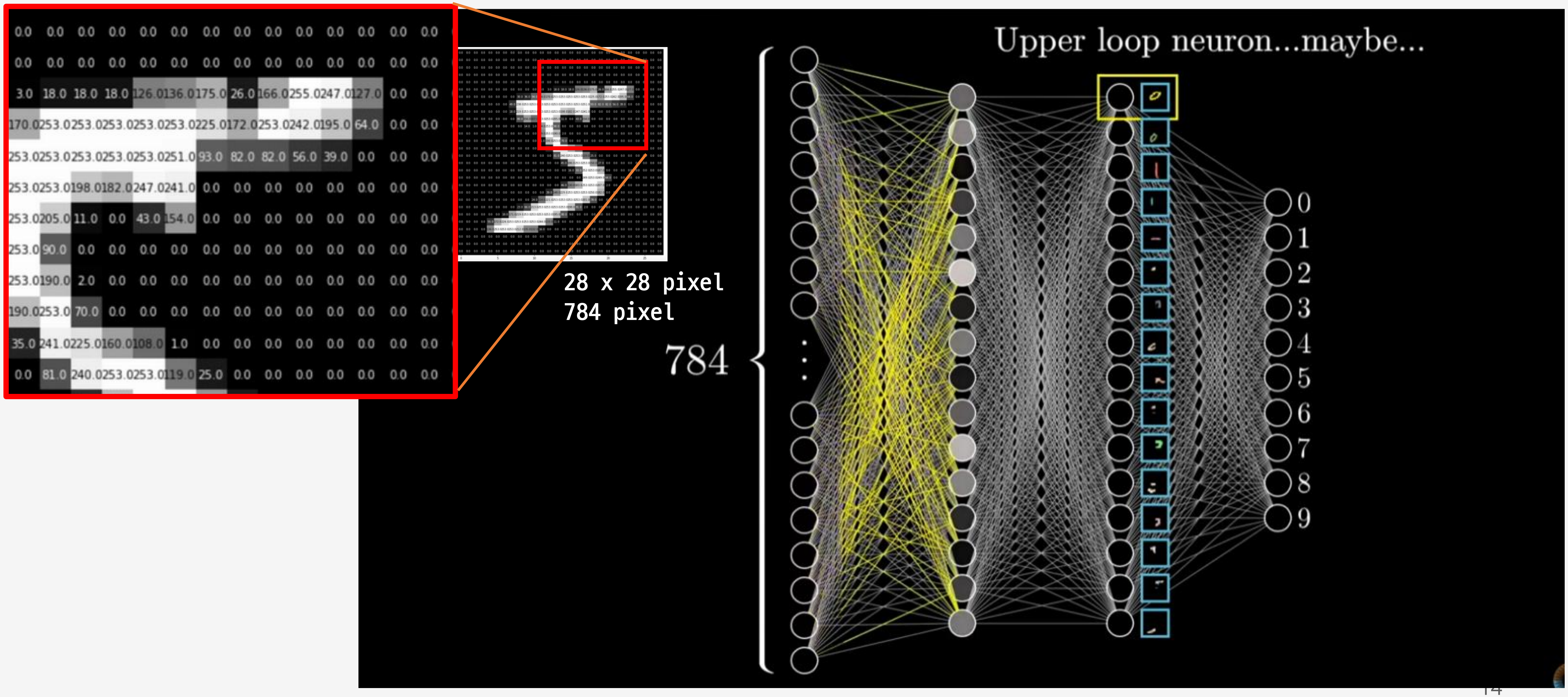
“Morpheus: The Matrix is everywhere. It is all around us. Even now, in this very room. You can see it when you look out your window or when you turn on your television. You can feel it when you go to work... when you go to church... when you pay your taxes. It is the world that has been pulled over your eyes to blind you from the truth.

Neo: What truth?

Morpheus: That you are a slave, Neo. Like everyone else you were born into bondage. Into a prison that you cannot taste or see or touch. A prison for your mind.”

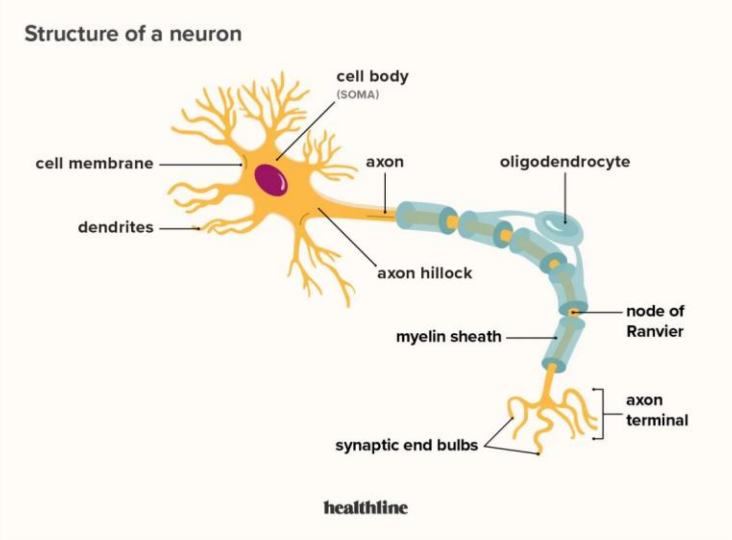
— Lana Wachowski, The Matrix: The Shooting Script

What is a deep learning?



What is a neural network?(start= 3 min, end = 5 min)

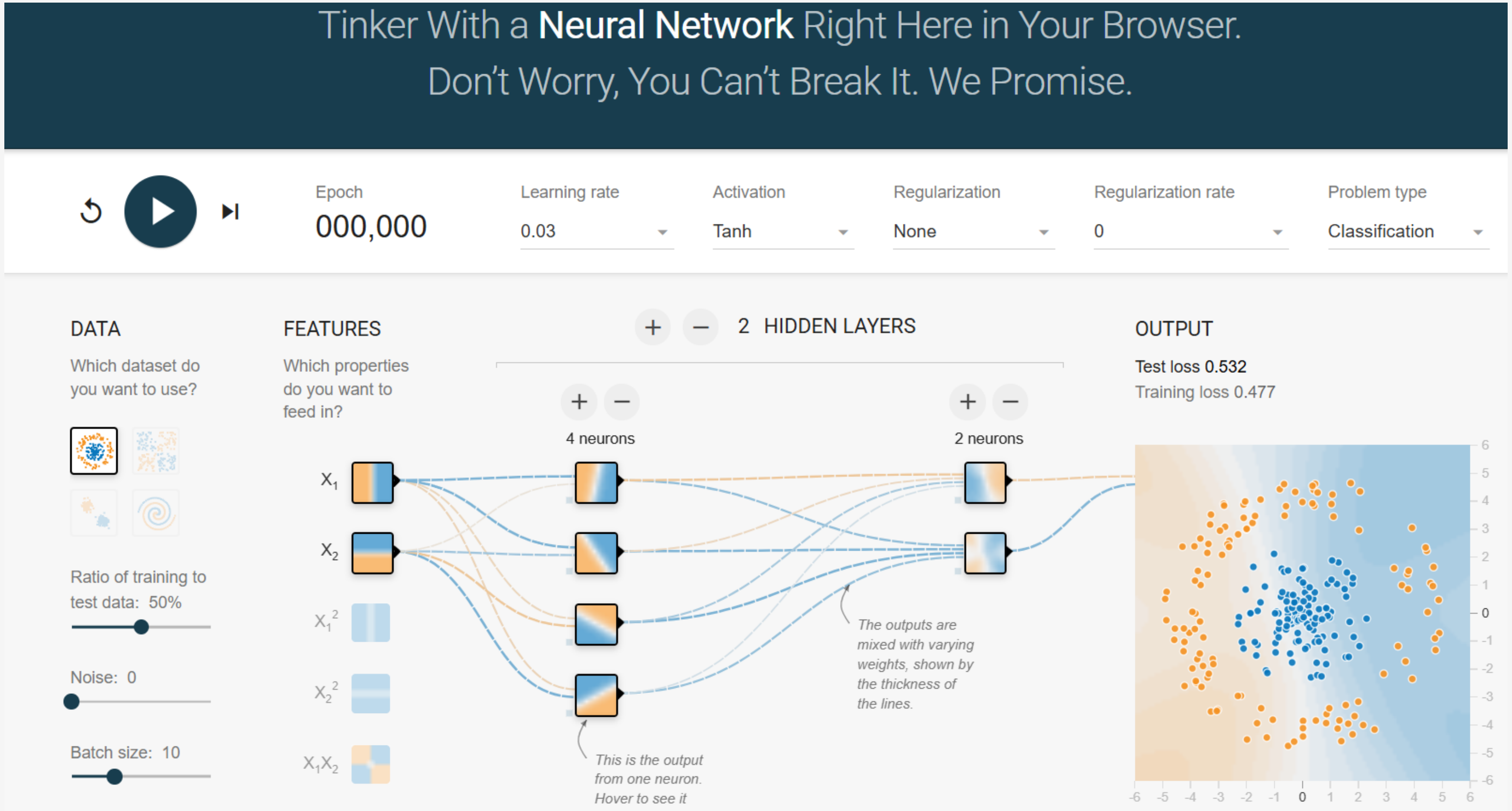
Human Neurons ?



https://unsplash.com/

실습하기

https://playground.tensorflow.org/

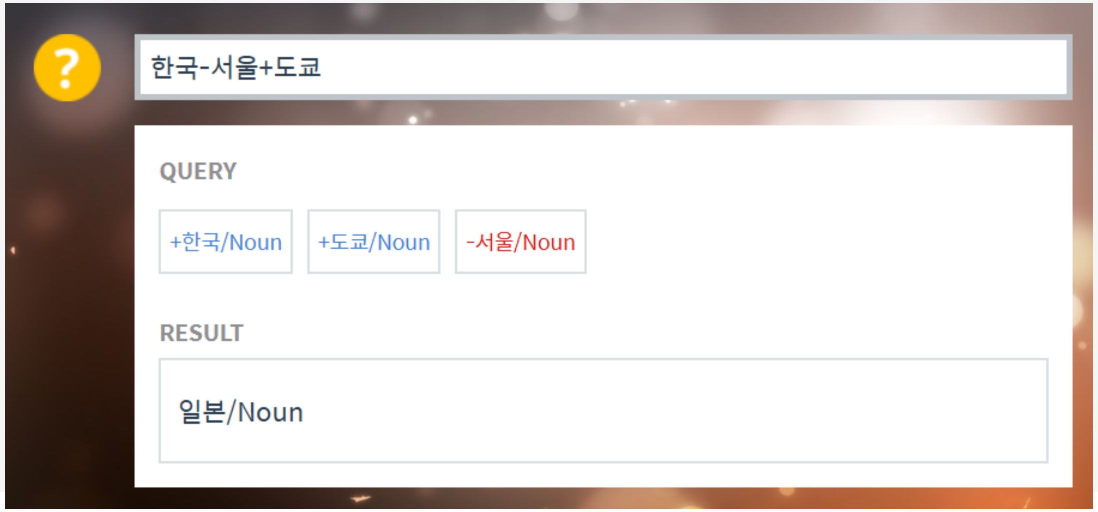


실습하기

Word Embedding

	have	the	bards	who	preceded	me	left	any	theme	unsung
0	-1.621	-0.634	-0.324	-1.357	-0.261	4.632	1.236	-0.046	-1.518	-0.082
1	-1.401	0.683	-0.114	1.891	0.544	-1.487	1.247	-0.408	0.202	-0.022
	…	…	…	…	…	…	…	…	…	…
49	-0.647	0.200	-0.203	1.573	-0.520	-0.355	-1.491	1.325	2.550	-0.010

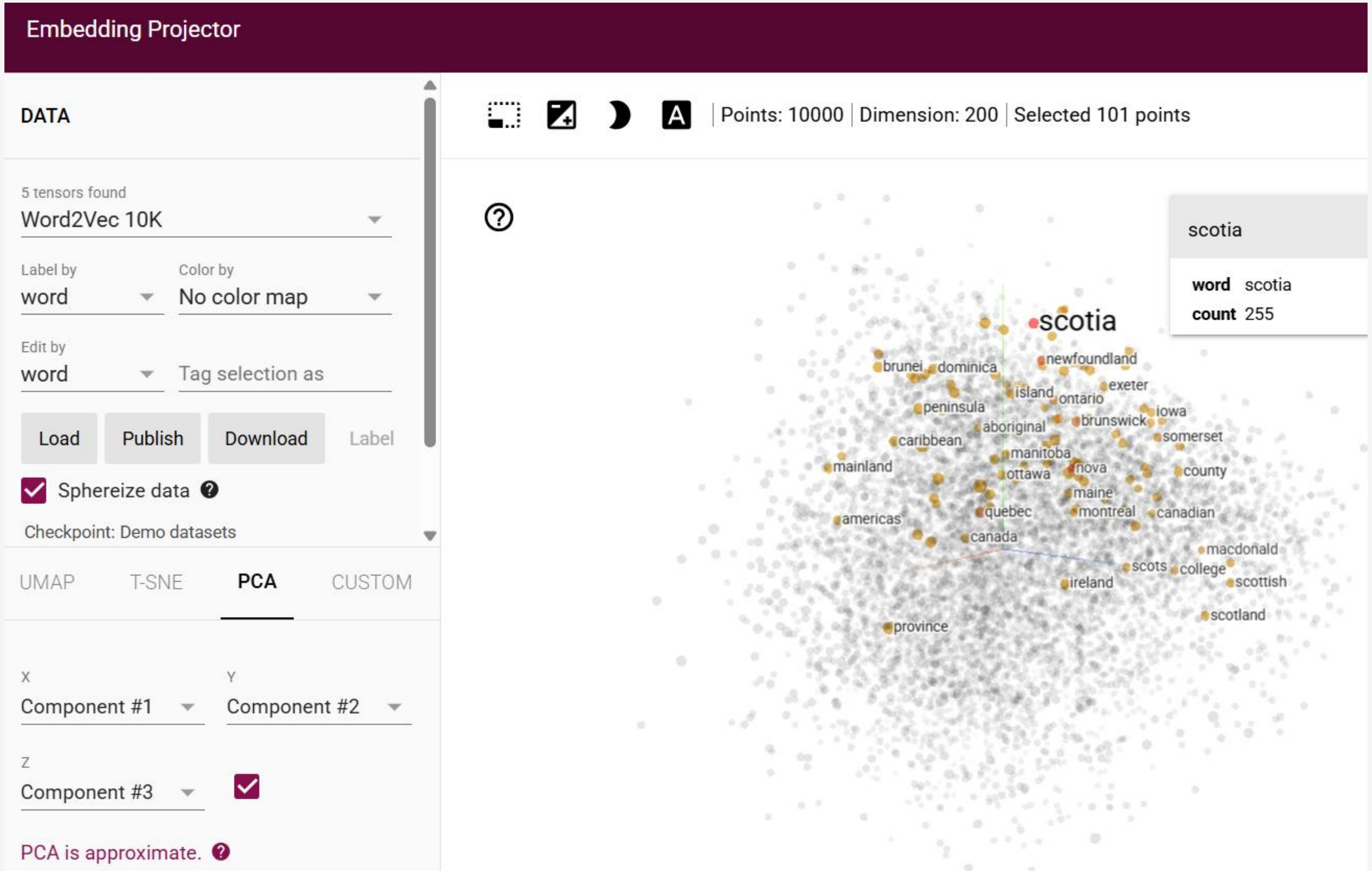
https://word2vec.kr/search/



한국-프랑스+파리

실습하기

tensorflow_projector



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대규모 언어 모델 (Large Language Model, LLM)

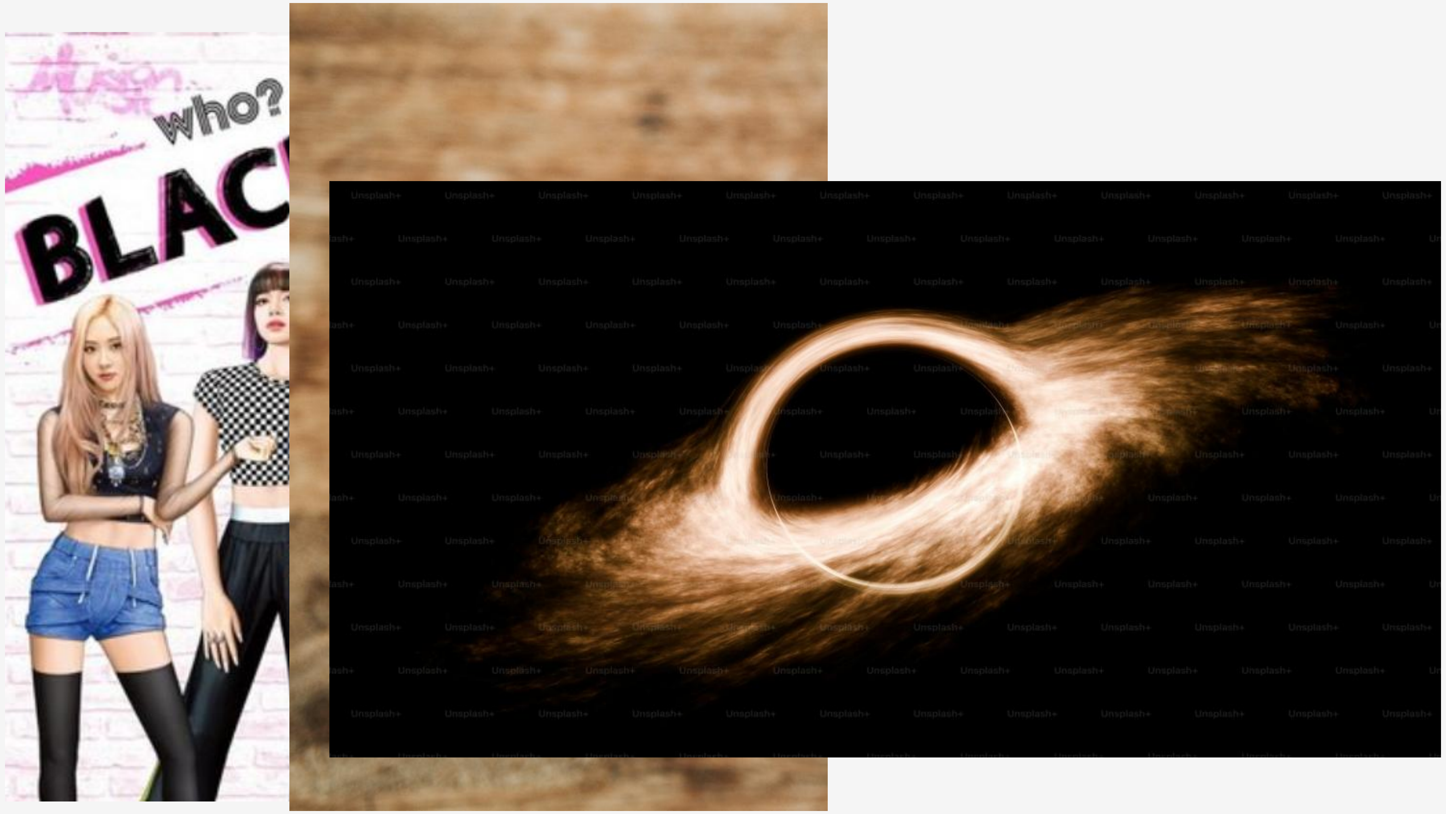
데이터를 학습한 딥러닝은 이전 데이터(단어, 픽셀 등)들을 기반으로 다음 데이터(단어, 픽셀)를 확률적으로 예측하여 텍스트, 이미지, 음악, 코드 등 새로운 콘텐츠를 생성(Generative)하는 AI

블랙 ?

핑크

커피

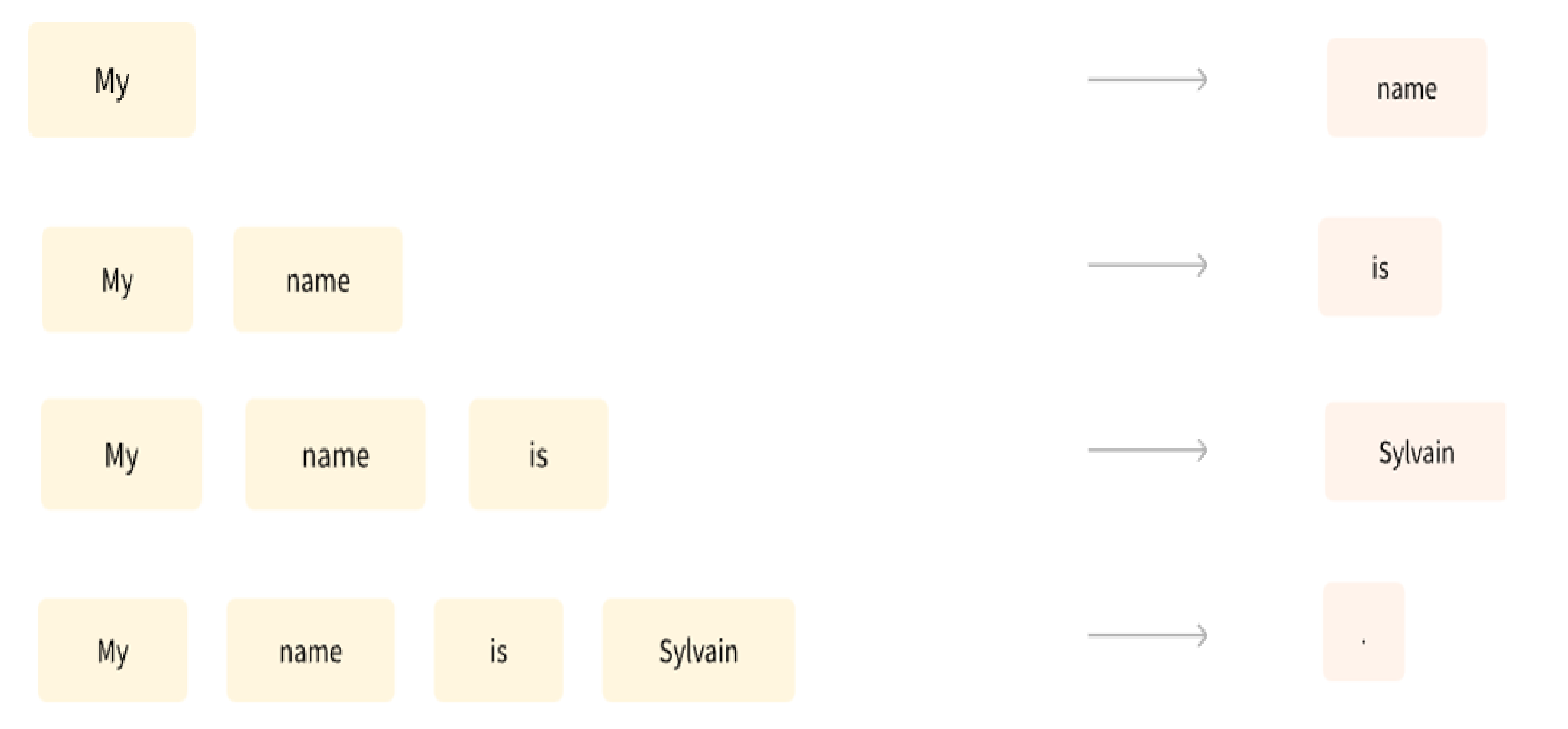
홀

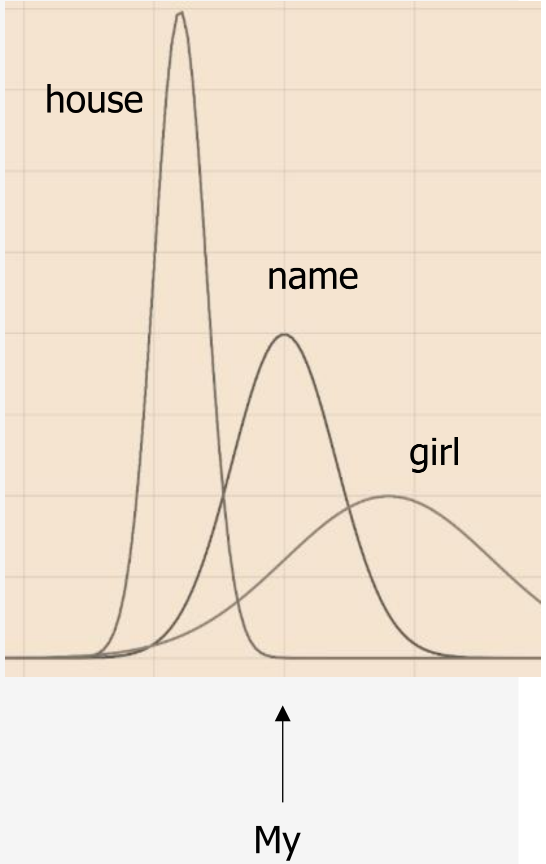


대규모 언어 모델 (Large Language Model, LLM)

확률적 언어모델

데이터를 학습한 딥러닝은 이전 단어들을 기반으로 다음 단어를 확률적으로 예측하여 말을 이어가는 끝 말 이어가기 끝판 왕 → **LLM(Large Language Model)**

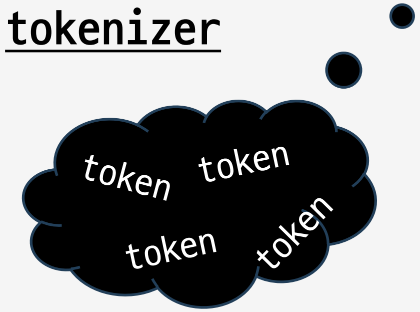


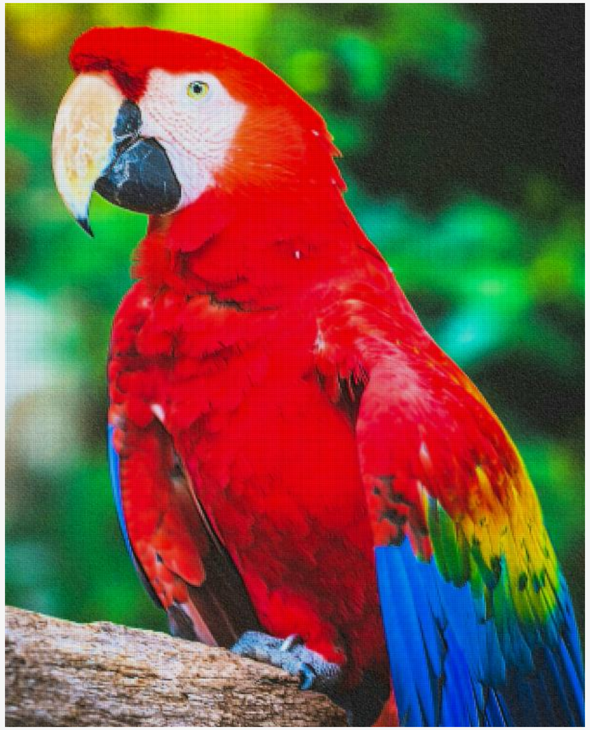


https://huggingface.co/learn/nlp-course/chapter1/4?fw=pt

확률적 언어모델을 가진 앵무새(Generative AI, LLM)

<u>tokenizer</u>





- 수십억 개의 단어를 학습하여 사람처럼 자연스럽게 언어를 생성하는 인공지능
- 신경망(Neural Network) 기반으로 문맥을 이해하고 논리적 답변 생성 가능
- 대표 모델: GPT-4, Claude, Gemini, LLaMA 등

LLM과 인간과 같은 감정으로 소통하진 마라

- ❖ **Gary Marcus:** "LLMs are fluent bullshitters."
  - ❖ 언어적 유창성은 뛰어나나 사실 검증 메커니즘이 없음
- ❖ **Yejin Choi:** "Commonsense missing, therefore bullshit-prone“
  - ❖ 세상에 대한 상식 없이 말하기 때문에 자주 현실과 동떨어짐
- ❖ 코마에서 깨어난 신입사원 아인슈타인

AI를 두려워하지 말고
도구로써 잘 이용하자

대규모 언어 모델 (Large Language Model, LLM)

- ❖ 수십억 개의 단어를 학습하여 사람처럼 자연스럽게 언어를 생성하는 인공지능
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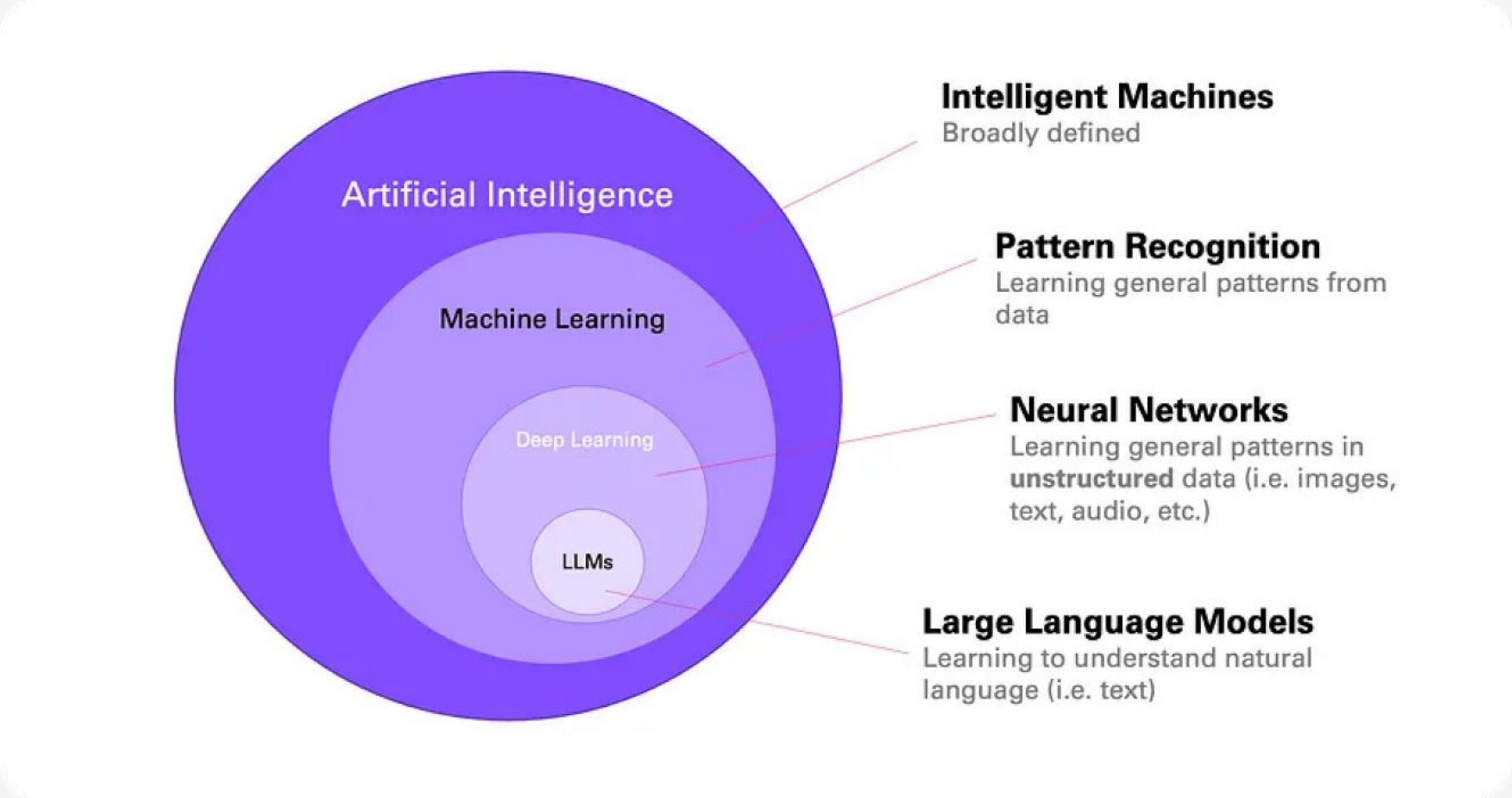
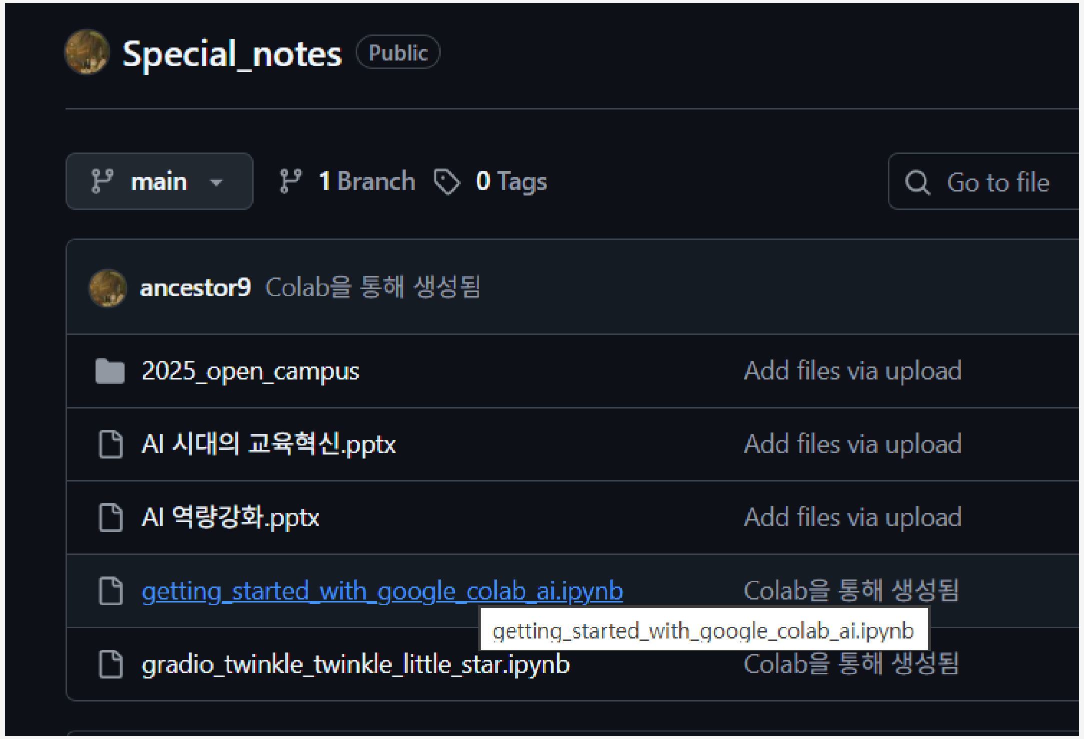


실습하기

제미나이 AI 실습 코드(colab)

1. 텍스트 생성(Generate text)
2. 번역하기(Translate languages)
3. 소설 만들기(Write creative content)
4. 보이스피싱 탐지(Categorize text)

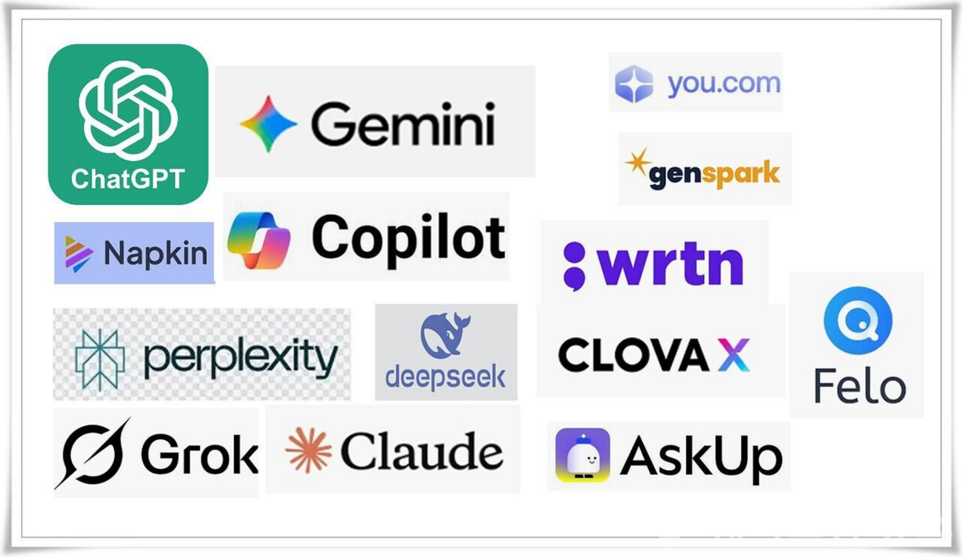


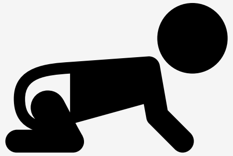
To be or not to be

AI(Artificial Intelligence) < IA (Intelligence Augmentation)

Intelligence

Augmentation





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실습하기

제미나이 AI 실습하는 코드(colab_python)

1. 텍스트 생성(Generate text)
2. 번역하기(Translate languages)
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↓

4가지 서비스를 친구에게 휴대폰이나
인터넷으로 제시하자 ! using gradio.app

gradio를 사용하여 4가지 기능을 만들어줘

LLM

언어 모델은 단어 시퀀스에 확률을 할당(assign) 하는 일을 하는 모델로 가장 자연스러운 단어 시퀀스를 찾아내어 예측하는 모델

LLMs → *Brain*

(plain language model)

❖ It talks.

→ you send it a message, it send text back.

❖ It just generate text and predict text. (It can’t check your email or it can’t search the Web…)

AI Agent

AI Agent is simply just a language model that can use tools that's running in a loop until it finishes a job.

Agent → *Hands*

(GIVE THIS MODEL 2 NEW ABILITIES)

❖ 1st Ability

→ take actions in the real world (Tool-calling)

❖ 2nd Ability

→ to keep going on its own step by step,
instead of stopping after just a single reply

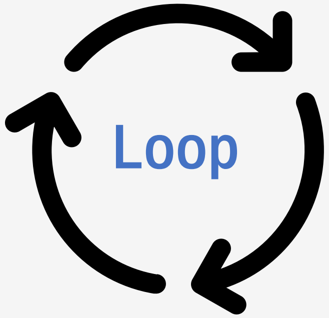
AI Agent

1 THINK

SYSTEM PROMPT & GOAL

4 THINK AGAIN

THE MODEL LOOKS AT THE RESULT AND DECIDES WHAT TO DO NEXT



2 ACT

MODEL DECIDES TO USE A TOOL AND SENDS THE ACTION

3 OBSERVE

CODE RUNS THE TOOL AND RETURN THE RESULT. THE MODEL OBSERVE THE RESULT

Tool-Calling

TOOL-CALLING

(The Hands)

- ❖ It generates text that tells whatever software it's working inside of to call a tool.

Tool-Calling

Context Window

(Short-term working Memory)

- ❖ The maximum amount of text it can process and remember at any single time during a conversation.

- ❖ Conversation history is tracked up to LLM’ Context Window.

Prompt, Message

System Prompt

- ❖ Is where you set the rules. So you tell the agent who it is, how to behave. General rules things that it should do.

User Messages

- ❖ The input from the person or the system.

Assistant Messages

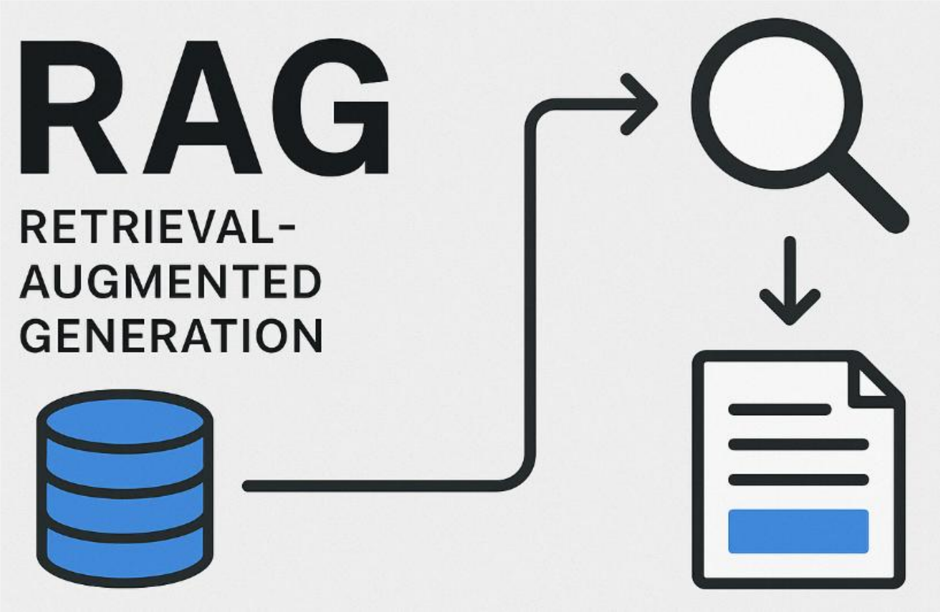
- ❖ What the model says back.

RAG

Conversation history can get quite large and it can actually not fit inside of the current context window → RAG

Embedding

Vector DB



How to build AI Agent

1 NO-CODE	Build an agent by filling out forms or dragging blocks around.
2 LOW-CODE	Write parts of the agent like a visual canvas.
3 AGENT HARNESSES	Install a complete powerful agent and customize it.
4 FULL CODE	Writing the entire thing in a programing language like Python.

How to build AI Agent

1 NO-CODE	Lindi, Gumloop, Stack AI, Relevance AI, Botpress, Genspark, Voiceflow
2 LOW-CODE	n8n, Flowise, Langflow, Dify, Activepieces, KNIME
3 AGENT HARNESSES	OpenClaw, HERMES AGENTS
4 FULL CODE	Python with Langchain.

Practice with google Agent SDK